

Applied Functional Analysis

THIRD EDITION

$$\begin{array}{ccc} U & \xrightarrow{B} & V' \\ R_U \downarrow & & \uparrow R_V \\ U' & \xleftarrow{B'} & V \sim V'' \end{array}$$

$$\begin{cases} u \in U \\ b(u, v) = l(v) \quad v \in V \end{cases}$$

J. Tinsley Oden

Leszek F. Demkowicz



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THIRD EDITION

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*In Memory of
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and
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Preface to the Third Edition

Applied Functional Analysis continues to provide a crash course for beginning graduate students with non-math majors, on the mathematical foundations of Functional Analysis, in the graduate program on *Computational and Applied Mathematics* at The University of Texas at Austin. Besides teaching the basics, through carefully selected examples, the text gradually introduces the student to the variational theory of partial differential equations, distributions, and Sobolev spaces, designed to prepare them for studying the mathematical aspects of numerical analysis and the mathematical theory of finite elements.

This third edition continues to serve both of these goals. This time, major revisions are included in [Chapters 5](#) and [6](#). The section on variational (weak) formulations is new. We consider the classical example of a diffusion-convection-reaction problem, reformulate it as a system of first order equations, and derive six different variational formulations. The formulations differ in their functional setting; each of them may lead to a separate Bubnov - or Petrov - Galerkin discretization and a Finite Element method that will converge in a different norm depending on the functional setting. We demonstrate then that the six formulations are simultaneously well posed. The proof leads naturally to the use of the Closed Range Theorems for both continuous and closed operators.

This basic step in the underlying theory motivated us to significantly expand the exposition on the Closed Range Theorem in Banach spaces in [Chapter 5](#), including a new section on the (full statement of the) Closed Range Theorem for closed operators.

Other minor changes include an expansion of the exposition on Schwartz's spaces of test functions. The topology of these spaces is not sequential, i.e., in general, continuity and sequential continuity of functions defined on $\mathcal{D}(\Omega)$ are not equivalent notions. Nevertheless, for linear functionals, the two notions coincide and, consequently, continuity and sequential continuity of distributions are shown to be equivalent notions. The extended discussion on $\mathcal{D}(\Omega)$ led to necessary changes and expansions of the earlier [Chapter 4](#).

The book continues to be used as a text for two one - semester courses. The first course covers the first four chapters and real analysis with a look at infinite dimensional spaces. The last two chapters are used in the second course teaching the actual theory of Banach and Hilbert spaces, motivated with variational formulations of boundary-value problems.

We thank a number of colleagues and students who made useful suggestions and read parts of the text during the preparation of the third edition. In particular, we mention Prof. Issam Louhichi and our students: Federico Fuentes, Sriram Nagaraj, Brendan Keith, and Stefan Henneking.

J. Tinsley Oden and Leszek F. Demkowicz

Austin, March 2017

Preface to the Second Edition

The rapid development of information and computer technologies and of the computational sciences has created an environment in which it is critically important to teach applicable mathematics in an interdisciplinary setting. The trend is best illustrated with the emergence of multiple graduate programs in applied and computational mathematics, and computational science and engineering across the nation and worldwide. With a finite number of curriculum hours and a multitude of new subjects, we are constantly faced with the dilemma of how to teach mathematics and which subjects to choose.

The main purpose of *Applied Functional Analysis for Science and Engineering* has been to provide a crash course for beginning graduate students with non math majors, on mathematical foundations leading to classical results in Functional Analysis. Indeed, it has served its purpose over the last decade of the graduate program on *Computational and Applied Mathematics* at The University of Texas. A more particular goal of the text has been to prepare the students to learn the variational theory of partial differential equations, distributions and Sobolev spaces and numerical analysis with an emphasis on finite element methods.

This second edition continues to serve both of these goals. We have kept the original structure of the book, resisting temptation of adding too many new topics. Instead, we have revised many of the original examples and added new ones, reflecting very often our own research experience and perspectives. In this revised edition, we start each chapter with an extensive introduction and conclude it with a summary and historical comments referring frequently to other sources. The number of exercises has been significantly increased and we are pleased to provide a *solution manual*. Problems provided in the text may be solved in many different ways, but the solutions presented are consistent with the style and philosophy of the presentation.

Main revisions of the material include the following changes:

Chapter 1: The order of presentations of elementary logic and elementary set theory has been reversed.

The section on $\limsup$ and $\liminf$ has been completely revised with functions taking values in the extended set of real numbers in mind. We have complemented the exposition on elementary topology with a discussion on connected sets.

Chapter 2: A new section on elements of multilinear algebra and determinants and a presentation on the Singular Value Decomposition Theorem have been added.

Chapter 3: We have added an example of a Lebesgue non measurable set, a short discussion on probability and Bayesian Statistical Inference, and a short presentation on the Cauchy Principal Value and Hadamard Finite Part integrals.

Chapter 4: We have added a discussion on connected sets.

Chapter 5: The discussion on representation theorems for duals of L^p -spaces has been complemented with the Generalized (Integral) Minkowski Inequality.

The book attempts to teach the rigor of logic and systematical, mathematical thinking. What makes it different from other mathematical texts is the large number of illustrative examples and comments. Engineering and science students come with a very practical attitude, and have to be constantly motivated and guided into appreciating the value and importance of mathematical rigor and the precision of thought that it provides. Nevertheless, the class in which the book has been used focuses on teaching how to prove theorems and prepares the students for further study of more advanced mathematical topics. The acquired ability to formulate research questions in a mathematically rigorous way has had a tremendous impact on our graduates and, we believe, it has been the best measure of the success of the text.

The book has been used as a text for a rather intensive two-semester course. The first semester focuses on real analysis with attention to infinite-dimensional settings, and it covers the first four chapters, culminating with the Banach Fixed Point Theorem. The second semester covers the actual Functional Analysis topics presented in [Chapters 5](#) and [6](#).

We thank a number of students and colleagues who made useful suggestions and read parts of the text during the preparation of the second edition: Tan Bui, Jessie Chan, Paolo Gatto, Antti Niemi, Frederick Qiu, Nathan Roberts, Jamie Wright, and Jeff Zitelli.

We thank James Goertz for helping with typing of the text.

J. Tinsley Oden and Leszek F. Demkowicz

Austin, September 2009

Preface to the First Edition

Worldwide, in many institutions of higher learning, there has emerged in recent years a variety of new academic programs designed to promote interdisciplinary education and research in applied and computational mathematics. These programs, advanced under various labels such as computational and applied mathematics, mathematical sciences, applied mathematics, and the like, are created to pull together several areas of mathematics, computer science, and engineering and science which underpin the broad subjects of mathematical modeling and computer simulation. In all such programs, it is necessary to bring students of science and engineering quickly to within reach of modern mathematical tools, to provide them with the precision and organization of thought intrinsic to mathematics, and to acquaint them with the fundamental concepts and theorems which form the foundation of mathematical analysis and mathematical modeling. These are among the goals of the present text.

This book, which is the outgrowth of notes used by the authors for over a decade, is designed for a course for beginning graduate students in computational and applied mathematics who enter the subject with backgrounds in engineering and science. The course purports to cover in a connected and unified manner an introduction to the topics in functional analysis important in mathematical modeling and computer simulation; particularly, the course lays the foundation for further work in partial differential equations, approximation theory, numerical mathematics, control theory, mathematical physics, and related subjects.

Prerequisites for the course for which this book is written are not extensive. The student with the usual background in calculus, ordinary differential equations, introductory matrix theory, and, perhaps, some background in applied advanced calculus typical of courses in engineering mathematics or introductory mathematical physics should find much of the book a logical and, we hope, exciting extension and abstraction of his knowledge of these subjects.

It is characteristic of such courses that they be paradoxical, in a sense, because on the one hand they presume to develop the foundations of algebra and analysis from the first principles, without appeal to any previous prejudices toward mathematical methods; but at the same time, they call upon undergraduate mathematical ideas repeatedly as examples or as illustrations of purpose of the abstractions and extensions afforded by the abstract theory. The present treatment is no exception.

We begin with an introduction to elementary set theoretics, logic, and general abstract algebra, and with an introduction to real analysis in [Chapter 1](#). [Chapter 2](#) is devoted to linear algebra in both finite and infinite dimensions. These two chapters could be skipped by many readers who have an undergraduate background in mathematics. For engineering graduate students, the material is often new and should be covered. We have provided numerous examples throughout the book to illustrate concepts, and many of these, again, draw

from undergraduate calculus, matrix theory, and ordinary differential equations.

[Chapter 3](#) is devoted to measure theory and integration and [Chapter 4](#) covers topological and metric spaces. In these chapters, the reader encounters the fundamentals of Lebesgue integration, L^p spaces, the Lebesgue Dominated Convergence Theorem, Fubini's Theorem, the notion of topologies, filters, open and closed sets, continuity, convergence, Baire categories, the contraction mapping principle, and various notions of compactness.

In [Chapter 5](#), all of the topological and algebraic notions covered in [Chapters 1–4](#) are brought together to study topological vector spaces and, particularly, Banach spaces. This chapter contains introductions to many fundamental concepts, including the theory of distributions, the Hahn–Banach Theorem and its corollaries, open mappings, closed operators, the Closed Graph Theorem, Banach Theorem, and the Closed Range Theorem. The main focus is on properties of linear operators on Banach spaces and, finally, the solution of linear equations.

[Chapter 6](#) is devoted to Hilbert spaces and to an introduction to the spectral theory of linear operators. There some applications to boundary-value problems of partial differential equations of mathematical physics are discussed in the context of the theory of linear operators on Hilbert spaces.

Depending upon the background of the entering students, the book may be used as a text for as many as three courses: [Chapters 1](#) and [2](#) provide a course on real analysis and linear algebra; [Chapters 3](#) and [4](#), a text on integration theory and metric spaces; and [Chapters 5](#) and [6](#) an introductory course on linear operators and Banach spaces. We have frequently taught all six chapters in a single semester course, but then we have been very selective of what topics were or were not taught. The material can be covered comfortably in two semesters, [Chapters 1–3](#) and, perhaps, part of [4](#) dealt with in the first semester and the remainder in the second.

As with all books, these volumes reflect the interests, prejudices, and experience of its authors. Our main interests lie in the theory and numerical analysis of boundary- and initial-value problems in engineering science and physics, and this is reflected in our choice of topics and in the organization of this work. We are fully aware, however, that the text also provides a foundation for a much broader range of studies and applications.

The book is very much based on the text with the same title by the first author and, indeed, can be considered as a new, extended, and revised version of it. It draws heavily from other monographs on the subject, listed in the References, as well as from various old personal lecture notes taken by the authors when they themselves were students. The second author would like especially to acknowledge the privilege of listening to unforgettable lectures of Prof. Stanisław Łojasiewicz at the Jagiellonian University in Cracow, from which much of the text on integration theory has been borrowed.

We wish to thank a number of students and colleagues who made useful suggestions and read parts of the text during the preparation of this work: Waldek Rachowicz, Andrzej Karafiat, Krzysztof Banaś, Tarek

Zohdi, and others. We thank Ms. Judith Caldwell for typing a majority of the text.

J. Tinsley Oden and Leszek F. Demkowicz

Austin, September 1995



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Preliminaries

Elementary Logic and Set Theory

1.1 Sets and Preliminary Notations, Number Sets

An axiomatic treatment of algebra, as with all mathematics, must begin with certain primitive concepts that are intuitively very simple but that may be impossible to define very precisely. Once these concepts have been agreed upon, true mathematics can begin—structure can be added, and a logical pattern of ideas, theorems, and consequences can be unraveled. Our aim here is to present a brief look at certain elementary, but essential, features of mathematics, and this must begin with an intuitive understanding of the concept of a *set*.

The term set is used to denote a collection, assemblage, or aggregate of objects. More precisely, a set is a plurality of objects that we treat as a single object. The objects that constitute a set are called the *members* or *elements* of the set. If a set contains a finite number of elements, we call it a *finite set*; if a set contains an infinity of elements, we call it an *infinite set*. A set that contains no elements at all is called an *empty*, *void*, or *null set* and is generally denoted $\emptyset$.

For convenience and conciseness in writing, we should also agree here on certain standard assumptions and notations. For example, any collection of sets we consider will be regarded as a collection of subsets of some mathematically well defined set in order to avoid notorious paradoxes concerned with the “set of all sets,” etc. The sets to be introduced here will always be well defined in the sense that it will be possible to determine if a given element is or is not a member of a given set. We will denote sets by uppercase Latin letters such as A , B , C , ... and elements of sets by lowercase Latin letters such as a , b , c , ... The symbol $\in$ will be used to denote membership of a set. For example, $a \in A$ means “the element a belongs to the set A ” or “ a is a member of A .” Similarly, a stroke through $\in$ negates membership; that is, $a \notin A$ means “ a does not belong to A .”

Usually various objects of one kind or another are collected to form a set because they share some common property. Indeed, the commonality or the characteristic of its elements serves to define the set itself. If set A has a small finite number of elements, the set can be defined simply by displaying all of its elements. For example, the set of natural (whole) numbers greater than 2 but less than 8 is written

$$A = \{3, 4, 5, 6, 7\}$$

However, if a set contains an infinity of elements, it is obvious that a more general method must be used to define the set. We shall adopt a rather widely used method: Suppose that every element of a set A has a certain property P ; then A is defined using the notation

$$A = \{a : a \text{ has property } P\}$$

Here a is understood to represent a typical member of A . For example, the finite set of whole numbers mentioned previously can be written

$$A = \{a : a \text{ is a natural number; } 2 < a < 8\}$$

Again, when confusion is likely, we shall simply write out in full the defining properties of certain sets.

Sets of primary importance in calculus are the number sets. These include:

- the set of *natural* (whole) *numbers*

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

- the set of *integers* (this notation honors Zermelo, a famous Italian mathematician who worked on number theory)

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

- the set of *rational numbers* (fractions)

$$\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{N} \right\}$$

- the set of *real numbers* $\mathbb{R}$
- the set of *complex numbers* $\mathbb{C}$

We do not attempt here to give either axiomatic or constructive definitions of these sets. Intuitively, once the notion of a natural number is adopted, $\mathbb{Z}$ may be constructed by adding zero and negative numbers, and $\mathbb{Q}$ is the set of fractions with integer numerators and natural (in particular different from zero) denominators. The real numbers may be identified with their decimal representations, and complex numbers may be viewed as pairs of real numbers with a specially defined multiplication.

The block symbols introduced above will be used hereafter to denote the number sets.

Subsets and Equality of Sets. If A and B are two sets, A is said to be a *subset* of B if and only if *every* element of A is also an element of B . The subset property is indicated by the symbolism

$$A \subset B$$

which is read “ A is a subset of B ” or, more frequently, “ A is contained in B .” Alternately, the notation $B \supset A$ is sometimes used to indicate that “ B contains A ” or “ B is a ‘superset’ of A .”

It is clear from this definition that every set A is a subset of itself. To describe subsets of a given set B that do not coincide with B , we use the idea of *proper subsets*; a set A is a proper subset of B if and only if A is a subset of B and B contains one or more elements that do not belong to A . Occasionally, to emphasize that A is a subset of B but possibly not a proper subset, we may write $A \subseteq B$ or $B \supseteq A$.

We are now ready to describe what is meant by equality of two sets. It is tempting to say that two sets are “equal” if they simply contain the same elements, but this is a little too imprecise to be of much value in proofs of certain set relations to be described subsequently. Rather, we use the equivalent idea that equal sets must contain each other; two sets A and B are said to be *equal* if and only if $A \subset B$ and $B \subset A$. If A is equal to B , we write

$$A = B$$

In general, to prove equality of two sets A and B , we first select a typical member of A and show that it belongs to the set B . Then, by definition, $A \subset B$. We then select a typical member of B and show that it also belongs to A , so that $B \subset A$. The equality of A and B then follows from the definition.

Exercises

Exercise 1.1.1 If $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ denotes the set of all integers and $\mathbb{N} = \{1, 2, 3, \dots\}$ the set of all natural numbers, exhibit the following sets in the form $A = \{a, b, c, \dots\}$:

- (i) $\{x \in \mathbb{Z} : x^2 - 2x + 1 = 0\}$
- (ii) $\{x \in \mathbb{Z} : 4 \leq x \leq 10\}$
- (iii) $\{x \in \mathbb{N} : x^2 < 10\}$

1.2 Level One Logic

Statements. Before we turn to more complicated notions like relations or functions, we would do well to examine briefly some elementary concepts in logic so that we may have some idea about the meaning of a proof. We are not interested here in examining the foundations of mathematics, but in formalizing certain types of thinking people have used for centuries to derive meaningful conclusions from certain premises. Millennia ago, the ancient Greeks learned that a deductive argument must start somewhere. In other words, certain statements, called *axioms*, are *assumed* to be true and then, by reasonable arguments, new “true” statements are derived. The notion of “truth” in mathematics may thus have nothing to do with concepts of “truth” (whatever the term may mean) discussed by philosophers. It is merely the starting point of an exercise in which new true statements are derived from old ones by certain fixed rules of logic. We expect

that there is general agreement among knowledgeable specialists that this starting point is acceptable and that the consequences of the choices of truth agree with our experiences.

Typically, a branch of mathematics is constructed in the following way. A small number of statements called *axioms* is assumed to be true. To signify this, we may assign the letter “t” (true) to them. Then there are various ways to construct new statements, and some specific rules are prescribed to assign the value “t” or “f” (false) to them. Each of the new statements must be assigned only one of the two values. In other words, no situation can be accepted in which a statement could be simultaneously true and false. If this happens, it will mean that the set of axioms is *inconsistent* and the whole theory should be abandoned (at least from the mathematical point of view; there are many inconsistent theories in engineering practice and they are still in operation).

For a consistent set of axioms, the statements bearing the “t” value are called *theorems*, *lemmas*, *corollaries*, and *propositions*. Though many inconsistencies in using these words are encountered, the following rules may be suggested:

- a *theorem* is an important true statement;
- a *lemma* is a true statement, serving, however, as an auxiliary tool to prove a certain theorem or theorems;
- a *proposition* is (in fact) a theorem which is not important enough to be called a *theorem*. This suggests that the name *theorem* be used rather rarely to emphasize especially important key results;
- finally, a *corollary* is a true statement, derived as an immediate consequence of a theorem or proposition with little extra effort.

Lowercase letters will be used to denote statements. Typically, letters p , q , r , and s are preferred. Recall once again that a statement p is a sentence for which only one of the two values “true” or “false” can be assigned.

Statement Operations, Truth Tables. In the following, we shall list the fundamental operations on statements that allow us to construct new statements, and we shall specify precisely the way to assign the “true” and “false” values to those new statements.

Negation: $\sim q$, to be read: not q

If $p = \sim q$ then p and q always bear opposite values; p is false when q is true and, conversely, if q is false then p is true. Assigning value 1 for “true” and 0 for “false,” we may illustrate this rule using the so-called truth table:

q	$\sim q$
1	0
0	1

Alternative: $p \vee q$, to be read: p or q

The alternative $r = p \vee q$ is true whenever at least one of the two component statements p or q is true. In other words, r is false only when both p and q are false. Again we can use the truth table to illustrate the definition:

p	q	$p \vee q$
1	1	1
1	0	1
0	1	1
0	0	0

Note in particular the non-exclusive character of the alternative. The fact that $p \vee q$ is true does not indicate that only one of the two statements p or q is true; they *both* may be true. This is somewhat in conflict with the everyday use of the word “or.”

Conjunction: $p \wedge q$, to be read: p and q

The conjunction $p \wedge q$ is true only if both p and q are true. We have the following truth table:

p	q	$p \wedge q$
1	1	1
1	0	0
0	1	0
0	0	0

Implication: $p \Rightarrow q$, to be read in one of the following ways:

- p implies q
- q if p
- q follows from p
- if p then q
- p is a sufficient condition for q
- q is a necessary condition for p

It is somewhat confusing, but all these sentences mean exactly the same thing. The truth table for implication is as follows:

p	q	$p \Rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1

Thus, the implication $p \Rightarrow q$ is false only when “true” implies “false.” Surprisingly, a false statement may imply a true one and the implication is still considered to be true.

Equivalence: $p \Leftrightarrow q$, to be read: p is equivalent to q .

The truth table is as follows:

p	q	$p \Leftrightarrow q$
1	1	1
1	0	0
0	1	0
0	0	1

Thus the equivalence $p \Leftrightarrow q$ is true (as expected) when both p and q are simultaneously true or false.

All theorems, propositions, etc., are formulated in the form of an implication or an equivalence. Notice that in proving a theorem in the form of implication $p \Rightarrow q$, we typically assume that p is true and attempt to show that q must be true. We do not need to check what will happen if p is false. No matter which value q takes on, the whole implication will be true.

Tautologies. Using the five operations on statements, we may build new combined operations and new statements. Some of them always turn out to be true no matter which values are taken on by the initial statements. Such a statement is called in logic a *tautology*.

As an example, let us study the fundamental statement known as one of *De Morgan's Laws* showing the relation between the negation, alternative, and conjunction.

$$\sim (p \vee q) \Leftrightarrow (\sim p) \wedge (\sim q)$$

One of the very convenient ways to prove that this statement is a tautology is to use truth tables.

We begin by noticing that the tautology involves two elementary statements p and q . As both p and q can take two logical values, 0 (false) or 1 (true), we have to consider a total of $2^2 = 4$ cases. We begin by organizing these cases using the lexicographic ordering (same as in a car's odometer):

p	q
0	0
0	1
1	0
1	1

It is convenient to write down these logical values directly underneath symbols p and q in the statement:

$$\begin{array}{cccc} \sim (p \vee q) & \Leftrightarrow & (\sim p) & \wedge & (\sim q) \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{array}$$

The first logical operations made are the negations on the right-hand side and the alternative on the left-hand side. We use the truth table for the negation and the alternative to fill in the proper values:

$$\sim (p \vee q) \Leftrightarrow (\sim p) \wedge (\sim q)$$

0	0	0	1	0	1	0
0	1	1	1	0	0	1
1	1	0	0	1	1	0
1	1	1	0	1	0	1

The next logical operations are the negation on the left-hand side and the conjunction on the right-hand side. We use the truth tables for the negation and conjunction to fill in the corresponding values:

$$\sim (p \vee q) \Leftrightarrow (\sim p) \wedge (\sim q)$$

1	0	0	0	1	0	1	1	0
0	0	1	1	1	0	0	0	1
0	1	1	0	0	1	0	1	0
0	1	1	1	0	1	0	0	1

Finally, we use the truth table for the equivalence to find out the ultimate logical values for the statement:

$$\sim (p \vee q) \Leftrightarrow (\sim p) \wedge (\sim q)$$

1	0	0	0	1	1	0	1	1	0
0	0	1	1	1	1	0	0	0	1
0	1	1	0	1	0	1	0	1	0
0	1	1	1	1	0	1	0	0	1

The column underneath the equivalence symbol $\Leftrightarrow$ contains only the truth values (1's), which prove that the statement is a tautology. Obviously, it is much easier to do this on a blackboard.

In textbooks, we usually present only the final step of the procedure. Our second example involves the fundamental statement showing the relation between the implication and equivalence operations:

$$(p \Leftrightarrow q) \Leftrightarrow ((p \Rightarrow q) \wedge (q \Rightarrow p))$$

The corresponding truth table looks as follows:

$$((p \Leftrightarrow q) \Leftrightarrow ((p \Rightarrow q) \wedge (q \Rightarrow p)))$$

0	1	0	1	0	1	0	1	0	1	0
0	0	1	1	0	1	1	0	1	0	0
1	0	0	1	1	0	0	0	0	1	1
1	1	1	1	1	1	1	1	1	1	1

The law just proven is very important in proving theorems. It says that whenever we have to prove a theorem in the form of the equivalence $p \Leftrightarrow q$, we need to show that both $p \Rightarrow q$ and $q \Rightarrow p$. The fact is commonly expressed by replacing the phrase “ p is equivalent to q ” with “ p is a necessary and sufficient condition for q .”

Another very important law, fundamental for the methodology of proving theorems, is as follows:

$$(p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$$

Again, the truth table method can be used to prove that this statement is always true (see [Exercise 1.2.2](#)). This law lays down the foundation for the so-called *proof by contradiction*. In order to prove that p implies q , we negate q and show that this implies $\sim p$.

Example 1.2.1

As an example of the proof by contradiction, we shall prove the following simple proposition:

If $n = k^2 + 1$, and k is natural number, then n cannot be a square of a natural number.

Assume, contrary to the hypothesis, that $n = \ell^2$. Thus $k^2 + 1 = \ell^2$ and, consequently,

$$1 = \ell^2 - k^2 = (\ell - k)(\ell + k)$$

a contradiction, since $\ell - k \neq \ell + k$ and 1 is divisible only by itself. $\square$

In practice, there is more than one assumption in a theorem; this means that statement p in the theorem $p \Rightarrow q$ is not a simple statement but rather a collection of many statements. Those include all of the theorems (true statements) of the theory being developed which are not necessarily listed as explicit assumptions. Consider for example the proposition:

$\sqrt{2}$ is not a rational number

It is somewhat confusing that this proposition is not in the form of an implication (nor equivalence). It looks to be just a single (negated) statement. In fact the proposition should be read as follows:

If all the results concerning the integers and the definition of rational numbers hold, then $\sqrt{2}$ is not a rational number.

We may proceed now with the proof as follows. Assume, to the contrary, that $\sqrt{2}$ is a rational number. Thus $\sqrt{2} = \frac{p}{q}$, where p and q are integers and may be assumed, without loss of generality (why?), to have no common divisor. Then $2 = \frac{p^2}{q^2}$, or $p^2 = 2q^2$. Thus p must be even. Then p^2 is divisible by 4, and hence q is even. But this means that 2 is a common divisor of p and q , a contradiction of the definition of rational numbers and the assumption that p and q have no common divisor.

Exercises

Exercise 1.2.1 Construct the truth table for *De Morgan's Law*:

$$\sim (p \wedge q) \Leftrightarrow ((\sim p) \vee (\sim q))$$

Exercise 1.2.2 Construct truth tables to prove the following tautologies:

$$(p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$$

$$\sim (p \Rightarrow q) \Leftrightarrow p \wedge \sim q$$

Exercise 1.2.3 Construct truth tables to prove the associative laws in logic:

$$p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$$

$$p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$$

1.3 Algebra of Sets

Set Operations. Some structure can be added to the rather loose idea of a set by defining a number of so-called set operations. We shall list several of these here. As a convenient conceptual aid, we also illustrate these operations by means of Venn diagrams in Fig. 1.1; there an abstract set is represented graphically by a closed region in the plane. In this figure, and in all of the definitions listed below, sets A , B , etc., are considered to be subsets of some fixed master set U called the *universal set*; the universal set contains all elements of a type under investigation.

Union. The *union* of two sets A and B is the set of all elements x that belong to A or B . The union of A and B is denoted by $A \cup B$ and, using the notation introduced previously,

$$A \cup B \stackrel{\text{def}}{=} \{x : x \in A \text{ or } x \in B\}$$

Thus an element in $A \cup B$ may belong to either A or B or to both A and B . The equality holds *by definition* which is emphasized by using the symbol $\stackrel{\text{def}}{=}$. Frequently, we replace symbol $\stackrel{\text{def}}{=}$ with a more compact and explicit notation “:=.” The colon on the *left* side of the equality sign indicates additionally that we are defining the quantity on the left.

Notice also that the definition of the union involves the logical operation of alternative. We can rewrite the definition using the symbol for alternative:

$$A \cup B \stackrel{\text{def}}{=} \{x : x \in A \vee x \in B\}$$

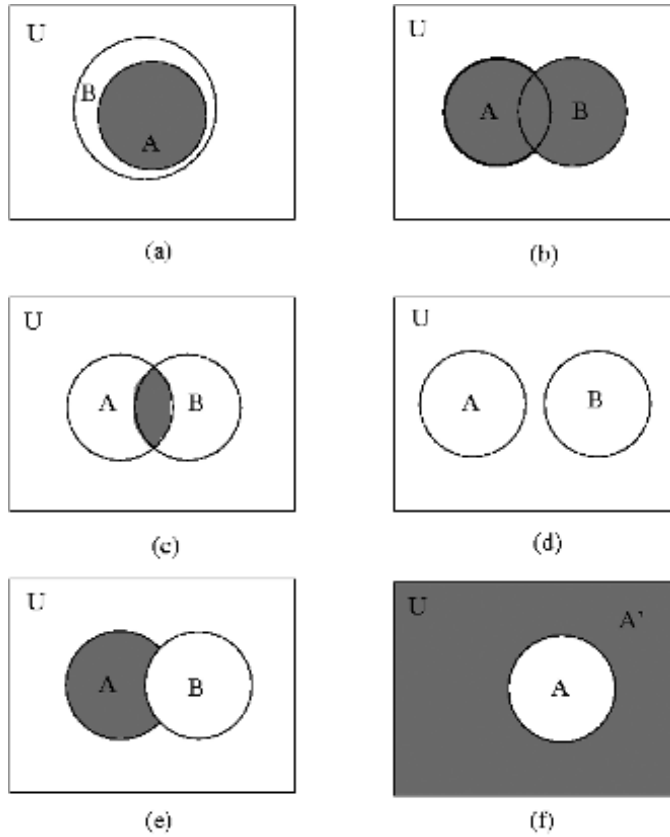
Equivalently, we can use the notion of logical equivalence to write:

$$x \in A \cup B \stackrel{\text{def}}{\Leftrightarrow} x \in A \vee x \in B$$

Again, the equivalence holds by definition. In practice, we limit the use of logical symbols and use verbal statements instead.

Intersection. The *intersection* of two sets A and B is the set of elements x that belong to both A and B . The symbolism $A \cap B$ is used to denote the intersection of A and B :

$$A \cap B \stackrel{\text{def}}{=} \{x : x \in A \text{ and } x \in B\}$$

**Figure 1.1**

Venn diagrams illustrating set relations and operations; a: $A \subset B$, b: $A \cup B$, c: $A \cap B$, d: $A \cap B = \emptyset$, e: $A - B$, f: A' .

Equivalently,

$$x \in A \cap B \quad \stackrel{\text{def}}{\Leftrightarrow} \quad x \in A \text{ and } x \in B$$

Disjoint Sets. Two sets A and B are *disjoint* if and only if they have no elements in common. Then their intersection is the empty set $\emptyset$ described earlier:

$$A \cap B = \emptyset$$

Difference. The *difference* of two sets A and B , denoted $A - B$, is the set of all elements that belong to A but not to B .

$$A - B \stackrel{\text{def}}{=} \{x : x \in A \text{ and } x \notin B\}$$

Equivalently,

$$x \in A - B \quad \stackrel{\text{def}}{\Leftrightarrow} \quad x \in A \text{ and } x \notin B$$

Complement. The *complement* of a set A (with respect to some universal set U), denoted by A' , is the set of elements which do not belong to A :

$$A' = \{x : x \in U \text{ and } x \notin A\}$$

In other words, $A' = U - A$ and $A' \cup A = U$. In particular, $U' = \emptyset$ and $\emptyset' = U$.

Example 1.3.1

Suppose U is the set of all lowercase Latin letters in the alphabet, A is the set of vowels ($A = \{a, e, i, o, u\}$), $B = \{c, d, e, i, r\}$, $C = \{x, y, z\}$. Then the following hold:

$$A' = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}$$

$$A \cup B = \{a, c, d, e, i, o, r, u\}$$

$$(A \cup B) \cup C = \{a, c, d, e, i, o, r, u, x, y, z\}$$

$$= A \cup (B \cup C)$$

$$A - B = \{a, o, u\}$$

$$B - A = \{c, d, r\}$$

$$A \cap B = \{e, i\} = B \cap A$$

$$A' \cap C = \{x, y, z\} = C \cap A'$$

$$U - ((A \cup B) \cup C) = \{b, f, g, h, j, k, l, m, n, p, q, s, t, v, w\}$$

□

Classes. We refer to sets whose elements are themselves sets as *classes*. Classes will be denoted by script letters $\mathcal{A}, \mathcal{B}, \mathcal{C}, \dots$, etc. For example, if A, B , and C are the sets

$$A = \{0, 1\}, \quad B = \{a, b, c\}, \quad C = \{4\}$$

the collection

$$\mathcal{A} = \{\{0, 1\}, \{a, b, c\}, \{4\}\}$$

is a class with elements A, B , and C .

Of particular interest is the *power set* or *power class* of a set A , denoted $\mathcal{P}(A)$. Based on the fact that a finite set with n elements has 2^n subsets, including $\emptyset$ and the set itself (see [Exercise 1.4.2](#)), $\mathcal{P}(A)$ is defined as the class of all subsets of A . Since there are 2^n sets in $\mathcal{P}(A)$, when A is finite we sometimes use the notation:

$$\mathcal{P}(A) = 2^A$$

Example 1.3.2

Suppose $A = \{1, 2, 3\}$. Then the power class $\mathcal{P}(A)$ contains $2^3 = 8$ sets:

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

It is necessary to distinguish between, e.g., element 1 and the single-element set $\{1\}$ (sometimes called *singleton*). Likewise, $\emptyset$ is the null set, but $\{\emptyset\}$ is a nonempty set with one element, that element being $\emptyset$. $\square$

Set Relations—Algebra of Sets The set operations described in the previous paragraph can be used to construct a sort of algebra of sets that is governed by a number of basic laws. We list several of these as follows:

Idempotent Laws

$$A \cup A = A; \quad A \cap A = A$$

Commutative Laws

$$A \cup B = B \cup A; \quad A \cap B = B \cap A$$

Associative Laws

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributive Laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Identity Laws

$$A \cup \emptyset = A; \quad A \cap U = A$$

$$A \cup U = U; \quad A \cap \emptyset = \emptyset$$

Complement Laws

$$A \cup A' = U; \quad A \cap A' = \emptyset$$

$$(A')' = A; \quad U' = \emptyset, \quad \emptyset' = U$$

There are also a number of special identities that often prove to be important. For example:

De Morgan's Laws

$$A - (B \cup C) = (A - B) \cap (A - C)$$

$$A - (B \cap C) = (A - B) \cup (A - C)$$

All of these so-called *laws* are merely theorems that can be proved by direct use of the definitions given in the preceding section and level-one logic tautologies.

Example 1.3.3

(Proof of Associative Laws)

We begin with the first law for the union of sets,

$$A \cup (B \cup C) = (A \cup B) \cup C$$

The law states the equality of two sets. We proceed in two steps. First, we will show that the left-hand side is contained in the right-hand side, and then that the right-hand side is also contained in the left-hand side. To show the first inclusion, we pick an arbitrary element $x \in A \cup (B \cup C)$. By definition of the union of two sets, this implies that

$$x \in A \text{ or } x \in (B \cup C)$$

By the same definition, this in turn implies

$$x \in A \text{ or } (x \in B \text{ or } x \in C)$$

If we identify now three logical statements,

$$\underbrace{x \in A}_p, \underbrace{x \in B}_q, \underbrace{x \in C}_r$$

the logical structure of the condition obtained so far is

$$p \vee (q \vee r)$$

It turns out that we have a corresponding *Associative Law for Alternative*:

$$p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$$

The law can be easily proved by using truth tables. By means of this law, we can replace our statement with an equivalent statement:

$$(x \in A \text{ or } x \in B) \text{ or } x \in C$$

Finally, recalling the definition of the union, we arrive at:

$$x \in (A \cup B) \cup C$$

We shall abbreviate the formalism by writing down all implications in a table, with the logical arguments listed on the right.

$$\begin{array}{ll}
 x \in A \cup (B \cup C) & \\
 \Downarrow & \text{definition of union} \\
 x \in A \text{ or } (x \in B \text{ or } x \in C) & \\
 \Downarrow & \text{tautology: } p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r \\
 (x \in A \text{ or } x \in B) \text{ or } x \in C & \\
 \Downarrow & \text{definition of union} \\
 x \in (A \cup B) \cup C &
 \end{array}$$

Finally, we notice that all of the implications can be reversed, i.e., in fact all statements are equivalent to each other:

$$\begin{array}{ll}
 x \in A \cup (B \cup C) & \\
 \Updownarrow & \text{definition of union} \\
 x \in A \text{ or } (x \in B \text{ or } x \in C) & \\
 \Updownarrow & \text{tautology: } p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r \\
 (x \in A \text{ or } x \in B) \text{ or } x \in C & \\
 \Updownarrow & \text{definition of union} \\
 x \in (A \cup B) \cup C &
 \end{array}$$

We have thus demonstrated that, conversely, each element from the right-hand side is also an element of the left-hand side. The two sets are therefore equal to each other. The second law is proved in a similar manner. $\square$

Example 1.3.4

(Proof of De Morgan's Laws)

We follow the same technique to obtain the following sequence of equivalent statements:

$$\begin{array}{ll}
 x \in A - (B \cup C) & \\
 \Updownarrow & \text{definition of difference of sets} \\
 x \in A \text{ and } x \notin (B \cup C) & \\
 \Updownarrow & x \notin D \Leftrightarrow \sim (x \in D) \\
 x \in A \text{ and } \sim (x \in B \cup C) & \\
 \Updownarrow & \text{definition of union} \\
 x \in A \text{ and } \sim (x \in B \vee x \in C) & \\
 \Updownarrow & \text{tautology: } p \wedge \sim (q \vee r) \Leftrightarrow (p \wedge \sim q) \wedge (p \wedge \sim r) \\
 (x \in A \text{ and } x \notin B) \text{ and } (x \in A \text{ and } x \notin C) & \\
 \Updownarrow & \text{definition of difference of sets} \\
 x \in (A - B) \text{ and } x \in (A - C) & \\
 \Updownarrow & \text{definition of intersection} \\
 x \in (A - B) \cap (A - C) &
 \end{array}$$

Notice that in the case of set A coinciding with the universal set U , the laws reduce to a simpler form expressed in terms of complements:

$$(B \cup C)' = B' \cap C'$$

$$(B \cap C)' = B' \cup C'$$

The second De Morgan's Law is proved in an analogous manner; see [Exercise 1.3.7](#). $\square$

The presented examples illustrate an intrinsic relation between the level-one logic tautologies and the algebra of sets. First of all, let us notice that we have implicitly used the operations on statements when defining the set operations. We have, for instance:

$$x \in A \cup B \Leftrightarrow (x \in A \vee x \in B)$$

$$x \in A \cap B \Leftrightarrow (x \in A \wedge x \in B)$$

$$x \in A' \Leftrightarrow \sim (x \in A)$$

Thus the notions like union, intersection, and complement of sets correspond to the notions of alternative, conjunction, and negation in logic. The situation became more evident when we used the laws of logic (tautologies) to prove the laws of algebra of sets. In fact, there is a one-to-one correspondence between laws of algebra of sets and laws of logic. The two theories express essentially the same algebraic facts. We will continue to illuminate this correspondence between sets and logic in subsequent sections.

Exercises

Exercise 1.3.1 Of 100 students polled at a certain university, 40 were enrolled in an engineering course, 50 in a mathematics course, and 64 in a physics course. Of these, only 3 were enrolled in all three subjects, 10 were enrolled only in mathematics and engineering, 35 were enrolled only in physics and mathematics, and 18 were enrolled only in engineering and physics.

- (i) How many students were enrolled only in mathematics?
- (ii) How many of the students were not enrolled in any of these three subjects?

Exercise 1.3.2 List all of the subsets of $A = \{1, 2, 3, 4\}$. Note: A and $\emptyset$ are considered to be subsets of A .

Exercise 1.3.3 Construct Venn diagrams to illustrate the idempotent, commutative, associative, distributive, and identity laws. Note: some of these are trivially illustrated.

Exercise 1.3.4 Construct Venn diagrams to illustrate De Morgan's Laws.

Exercise 1.3.5 Prove the distributive laws.

Exercise 1.3.6 Prove the identity laws.

Exercise 1.3.7 Prove the second of De Morgan's Laws.

Exercise 1.3.8 Prove that $(A - B) \cap B = \emptyset$.

Exercise 1.3.9 Prove that $B - A = B \cap A'$.

1.4 Level Two Logic

Open Statements, Quantifiers. Suppose that $S(x)$ is an expression which depends upon a variable x . One may think of variable x as the name of an unspecified object from a certain given set X . In general it is impossible to assign the “true” or “false” value to such an expression unless a specific value is substituted for x . If after such a substitution $S(x)$ becomes a statement, then $S(x)$ is called an *open statement*.

Example 1.4.1

Consider the expression:

$$x^2 > 3 \quad \text{with} \quad x \in \mathbb{N}$$

Then “ $x^2 > 3$ ” is an open statement which becomes true for x bigger than 1 and false for $x = 1$.
□

Thus, having an open statement $S(x)$ we may obtain a statement by substituting a specific variable from its domain X . We say that the *open statement has been closed by substitution*. Another way to close an open statement is to add to $S(x)$ one of the two so-called *quantifiers*:

$\forall x \in X$, to be read: for all x belonging to X , for every x in X , etc.

$\exists x \in X$, to be read: for some x belonging to X , there exists x in X such that, etc.

The first one is called the *universal quantifier* and the second the *existential quantifier*. Certainly by adding the universal quantifier to the open statement from [Example 1.4.1](#), we get the false statement:

$$\forall x \in \mathbb{N} \quad x^2 > 3$$

(every natural number, when squared is greater than 3), while by adding the existential qualifier we get the true statement:

$$\exists x \in \mathbb{N} \quad x^2 > 3$$

(there exists a natural number whose square is greater than 3).

Naturally, the quantifiers may be understood as generalizations of the alternative and conjunction. First of all, due to the associative law in logic (recall [Exercise 1.2.3](#)):

$$p \vee (q \vee r) \quad \Leftrightarrow \quad (p \vee q) \vee r$$

we may agree to define the alternative of these statements:

$$p \vee q \vee r$$

by either of the two statements above. Next, this can be generalized to the case of the alternative of the finite class of statements:

$$p_1 \vee p_2 \vee p_3 \vee \dots \vee p_N$$

Note that this statement is true whenever *there exists* a statement p_i , for some i , that is true. Thus, for finite sets $X = \{x_1, \dots, x_N\}$, the statement

$$\exists x \in X \quad S(x)$$

is equivalent to the alternative

$$S(x_1) \vee S(x_2) \vee \dots \vee S(x_N)$$

Similarly, the statement

$$\forall x \in X \quad S(x)$$

is equivalent to

$$S(x_1) \wedge S(x_2) \wedge \dots \wedge S(x_N)$$

Negation Rules for Quantifiers. We shall adopt the following *negation rule* for the universal quantifier:

$$\sim (\forall x \in X, S(x)) \quad \Leftrightarrow \quad \exists x \in X \quad \sim S(x)$$

Observe that this rule is consistent with De Morgan's Law:

$$\sim (p_1 \wedge p_2 \wedge \dots \wedge p_N) \quad \Leftrightarrow \quad (\sim p_1 \vee \sim p_2 \vee \dots \vee \sim p_N)$$

Substituting $\sim S(x)$ for $S(x)$ and negating both sides, we get the negation rule for the existential quantifier:

$$\sim (\exists x \in X, S(x)) \quad \Leftrightarrow \quad \forall x \in X \quad \sim S(x)$$

which again corresponds to the second De Morgan's Law:

$$\sim (p_1 \vee p_2 \vee \dots \vee p_N) \quad \Leftrightarrow \quad (\sim p_1 \wedge \sim p_2 \wedge \dots \wedge \sim p_N)$$

Principle of Mathematical Induction. Using the proof-by-contradiction concept and the negation rules for quantifiers, we can easily prove the *Principle of Mathematical Induction*. Let $T(n)$ be an open statement for $n \in \mathbb{N}$. Suppose that:

1. $T(1)$ (is true)
2. $T(k) \Rightarrow T(k+1) \quad \forall k \in \mathbb{N}$

Then,

$$T(n) \quad \forall n \quad (\text{is true})$$

PROOF Assume, to the contrary, that the statement $T(n) \forall n$ is not true. Then, by the negation rule, there exists a natural number, say k , such that $T(k)$ is false. This implies that the set

$$A = \{k \in \mathbb{N} : T(k) \text{ is false}\}$$

is not empty. Let l be the minimal element of A . Then $l \neq 1$ since, according to the assumption, $T(1)$ is true. Thus l must have a predecessor $l - 1$ for which $T(l - 1)$ holds. However, according to the second assumption, this implies that $T(l)$ is true as well: a contradiction. ■

It is easy to generalize the notion of open statements to more than one variable; for example:

$$S(x, y) \quad x \in X, y \in Y$$

Then the two negation rules may be used to construct more complicated negation rules for many variables, e.g.,

$$\sim (\forall x \in X \exists y \in Y S(x, y)) \Leftrightarrow \exists x \in X \forall y \in Y \sim S(x, y)$$

This is done by negating one quantifier at a time:

$$\begin{aligned} \sim (\forall x \in X \exists y \in Y S(x, y)) &\Leftrightarrow \sim (\forall x \in X (\exists y \in Y S(x, y))) \\ &\Leftrightarrow \exists x \in X \sim (\exists y \in Y S(x, y)) \\ &\Leftrightarrow \exists x \in X \forall y \in Y \sim S(x, y) \end{aligned}$$

We shall frequently use this type of technique throughout this book.

Exercises

Exercise 1.4.1 Use Mathematical Induction to derive and prove a formula for the sum of squares of the first n positive integers:

$$\sum_{i=1}^n i^2 = 1 + 2^2 + \dots + n^2$$

Exercise 1.4.2 Use mathematical induction to prove that the power set of a set U with n elements has 2^n elements:

$$\#U = n \Rightarrow \#\mathcal{P}(U) = 2^n$$

The hash symbol $\#$ replaces the phrase “number of elements of.”

1.5 Infinite Unions and Intersections

Unions and Intersections of Arbitrary Families of Sets. Notions of union and intersection of sets can be generalized to the case of arbitrary, possibly infinite families of sets. Let $\mathcal{A}$ be a class of sets A (possibly infinite). The *union* of sets from $\mathcal{A}$ is the set of all elements x that belong to *some* set from $\mathcal{A}$:

$$\bigcup_{A \in \mathcal{A}} A \stackrel{\text{def}}{=} \{x : \exists A \in \mathcal{A} : x \in A\}$$

Notice that in the notation above we have used the very elements of the family to “enumerate” or “label” themselves. This is a very convenient (and logically precise) notation and we will use it from time to time. Another possibility is to introduce an explicit index $\iota \in I$ to identify the family members:

$$\mathcal{A} = \{A_\iota : \iota \in I\}$$

We can use then an alternative notation to define the notion of the union:

$$\bigcup_{\iota \in I} A_\iota \stackrel{\text{def}}{=} \{x : \exists \iota \in I : x \in A_\iota\}$$

The ι indices on both sides are “dummy (summation) indices” and can be replaced with any other letter. By using the Greek letter ι in place of an integer index i , we emphasize that we are dealing with an arbitrary family.

In the same way we define the intersection of an arbitrary family of sets:

$$\bigcap_{A \in \mathcal{A}} A \stackrel{\text{def}}{=} \{x : \forall A \in \mathcal{A} \ x \in A\}$$

Traditionally, the universal quantifier is appended to the end of the statement:

$$\bigcap_{A \in \mathcal{A}} A \stackrel{\text{def}}{=} \{x : x \in A \ \forall A \in \mathcal{A}\}$$

Again, the same definition can be written out using explicit indexing:

$$\bigcap_{\iota \in I} A_\iota \stackrel{\text{def}}{=} \{x : x \in A_\iota \ \forall \iota \in I\}$$

As in the case of finite unions and intersections, we can also write these definitions in the following way:

$$\begin{aligned} x \in \bigcup_{\iota \in I} A_\iota &\stackrel{\text{def}}{\iff} \exists \iota \in I \ x \in A_\iota \\ x \in \bigcap_{\iota \in I} A_\iota &\stackrel{\text{def}}{\iff} x \in A_\iota \ \forall \iota \in I \end{aligned}$$

Example 1.5.1

Suppose $\mathbb{R}$ denotes the set of all real numbers and $\mathbb{R}^2$ the set of *ordered pairs* (x, y) of real numbers (we make these terms precise subsequently). Then the set $A_b = \{(x, y) \in \mathbb{R}^2 : y = bx\}$ is equivalent to the set of points on the straight line $y = bx$ in the Euclidean plane. The set of all such lines is the class

$$\mathcal{A} = \{A_b : b \in \mathbb{R}\}$$

In this case,

$$\bigcap_{b \in \mathbb{R}} A_b = \{(0, 0)\}$$

$$\bigcup_{b \in \mathbb{R}} A_b = \mathbb{R}^2 - \{(0, y) : |y| > 0\}$$

That is, the only point common to all members of the class is the origin $(0, 0)$, and the union of all such lines is the entire plane $\mathbb{R}^2$, excluding the y -axis, except the origin, since $b = \infty \notin \mathbb{R}$. $\square$

De Morgan's Laws can be generalized to the case of unions and intersections of arbitrary (in particular infinite) classes of sets:

$$A - \bigcup_{B \in \mathcal{B}} B = \bigcap_{B \in \mathcal{B}} (A - B)$$

$$A - \bigcap_{B \in \mathcal{B}} B = \bigcup_{B \in \mathcal{B}} (A - B)$$

When the universal set U is taken for A , we may use the notion of the complement of a set and write De Morgan's Laws in the more concise form

$$\left(\bigcup_{B \in \mathcal{B}} B \right)' = \bigcap_{B \in \mathcal{B}} B'$$

$$\left(\bigcap_{B \in \mathcal{B}} B \right)' = \bigcup_{B \in \mathcal{B}} B'$$

De Morgan's Laws express a *duality effect* between the notions of union and intersection of sets, and sometimes they are called the *duality laws*. They are a very effective tool in proving theorems.

The negation rules for quantifiers must be used when proving De Morgan's Laws for infinite unions and intersections. Indeed, the equality of sets

$$\left(\bigcap_{B \in \mathcal{B}} B \right)' = \bigcup_{B \in \mathcal{B}} B'$$

is equivalent to the statement

$$\sim (\forall B \in \mathcal{B}, x \in B) \Leftrightarrow \exists B \in \mathcal{B} \sim (x \in B)$$

and, similarly, the second law

$$\left(\bigcup_{B \in \mathcal{B}} B \right)' = \bigcap_{B \in \mathcal{B}} B'$$

corresponds to the second negation rule

$$\sim (\exists B \in \mathcal{B} x \in B) \Leftrightarrow \forall B \in \mathcal{B} \sim (x \in B)$$

Exercises

Exercise 1.5.1 Let $B(a, r)$ denote an open ball centered at a with radius r :

$$B(a, r) = \{x : d(x, a) < r\}$$

Here a, x are points in the Euclidean space and $d(x, a)$ denotes the (Euclidean) distance between the points. Similarly, let $\overline{B}(a, r)$ denote a closed ball centered at a with radius r :

$$\overline{B}(a, r) = \{x : d(x, a) \leq r\}$$

Notice that the open ball does not include the points on the sphere with radius r , whereas the closed ball does.

Determine the following infinite unions and intersections:

$$\begin{array}{cccc} \bigcup_{r < 1} B(a, r), & \bigcup_{r < 1} \overline{B}(a, r), & \bigcap_{r < 1} B(a, r), & \bigcap_{r < 1} \overline{B}(a, r), \\ \bigcup_{1 \leq r \leq 2} B(a, r), & \bigcup_{1 \leq r \leq 2} \overline{B}(a, r), & \bigcap_{1 \leq r \leq 2} B(a, r), & \bigcap_{1 \leq r \leq 2} \overline{B}(a, r) \end{array}$$

Relations

1.6 Cartesian Products, Relations

We are accustomed to the use of the term “relation” from elementary algebra. Intuitively, a *relation* must represent some sort of rule of correspondence between two or more objects; for example, “Bob is related to his brother Joe” or “real numbers are related to a scale on the x -axis.” One of the ways to make this concept more precise is to recall the notion of the open statement from the preceding section.

Suppose we are given an open statement of two variables:

$$R(x, y), \quad x \in A, \quad y \in B$$

We shall say that “ a is related to b ” and we write $a R b$ whenever $R(a, b)$ is true, i.e., upon the substitution $x = a$ and $y = b$, we get the true statement.

There is another equivalent way to introduce the notion of the relation by means of the set theory. First, we must introduce the idea of *ordered pairs* of mathematical objects and then the concept of the *product set*, or the *Cartesian product* of two sets.

Ordered Pairs. By an ordered pair (a, b) we shall mean the set $(a, b) = \{\{a\}, \{a, b\}\}$. Here a is called the first member of the pair and b the second member.

Cartesian Product. The Cartesian product of two sets A and B , denoted $A \times B$, is the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

We refer to the elements a and b as *components* of the pair (a, b) .

Two ordered pairs are equal if their respective components are equal, i.e.,

$$(x, y) = (a, b) \Leftrightarrow x = a \text{ and } y = b$$

Note that, in general,

$$A \times B \neq B \times A$$

More generally, if $A_1, A_2, \dots, A_k$ are k sets, we define the Cartesian product $A_1 \times A_2 \times \dots \times A_k$ to be the set of all ordered k -tuples $(a_1, a_2, \dots, a_k)$, where $a_i \in A_i, i = 1, 2, \dots, k$.

Example 1.6.1

Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$. Then

$$A \times B = \{(1, x), (1, y), (2, x), (2, y), (3, x), (3, y)\}$$

$$B \times A = \{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\}$$

□

Suppose now that we are given an open statement $R(x, y)$, $x \in A, y \in B$ and the corresponding relation R . With each such open statement $R(x, y)$ we may associate a subset of $A \times B$, denoted by R again, of the form:

$$R = \{(a, b) \in A \times B : a R b\} = \{(a, b) \in A \times B : R(a, b) \text{ holds}\}$$

In other words, with every relation R we may identify a subset of the Cartesian product $A \times B$ of all the pairs in which the first element is related to the second by R . Conversely, if we are given an arbitrary subset $R \subset A \times B$, then we may define the corresponding open statement as

$$R(x, y) = \{(x, y) \in R\}$$

which in turn implies that

$$a R b \Leftrightarrow (a, b) \in R$$

Thus there is the one-to-one correspondence between the two notions of relations which let us identify relations with subsets of the Cartesian products. We shall prefer this approach through most of this book.

More specifically, the relation $R \subseteq A \times B$ is called a *binary relation* since two sets A and B appear in the Cartesian product of which R is a subset. In general, we may define a “ k -ary” relation as a subset of $A_1 \times A_2 \times \dots \times A_k$.

The *domain* of a relation R is the set of all elements of A that are related by R to at least one element in B . We use the notation “ $\text{dom } R$ ” to denote the domain of a relation. Likewise, the *range* of R , denoted “ $\text{range } R$,” is the set of all elements of B to which at least one element of A is related by R . Thus:

$$\begin{aligned}\text{dom } R &= \{a : a \in A \text{ and } a R b \text{ for some } b \in B\} \\ \text{range } R &= \{b : b \in B \text{ and } a R b \text{ for some } a \in A\}\end{aligned}$$

We see that a relation, in much the same way as the common understanding of the word, is a rule that establishes an *association* of elements of a set A with those of another set B . Each element in the subset of A that is from R is associated by R with one or more elements in range R . The significance of particular relations can be quite varied; for example, the statement “Bob Smith is the father of John Smith” indicates a relation of “Bob Smith” to “John Smith,” the relation being “is the father of.” Other examples are cited below.

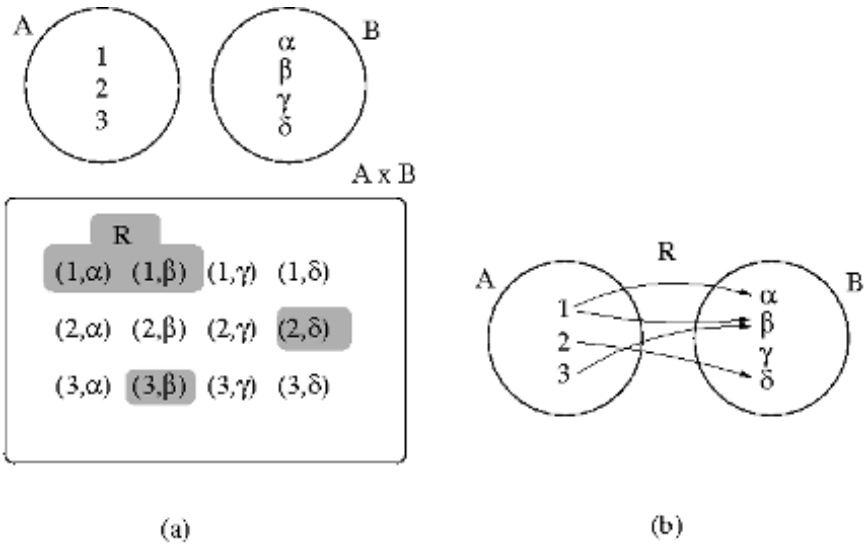


Figure 1.2
Graphical representation of a relation R from a set A to a set B .

Example 1.6.2

Let $A = \{1, 2, 3\}$ and $B = \{\alpha, \beta, \gamma, \delta\}$, and let R be the subset of $A \times B$ that consists of the pairs

$(1, \alpha), (1, \beta), (2, \delta), (3, \beta)$. Then

$$\text{dom } R = \{1, 2, 3\} = A$$

$$\text{range } R = \{\alpha, \beta, \delta\} \subset B$$

We see that R establishes a *multivalued correspondence* between elements of A and B . It is often instructive to represent relations such as this by diagrams; this particular example is indicated in Fig. 1.2 (a). Fig. 1.2 (b) depicts the relation R and “sending” or “mapping” certain elements of A into certain elements of B . $\square$

Example 1.6.3

Let $P = \{a, b, c, \dots\}$ be the set of all people in a certain school, and let $T = \{a, b, c\}$ denote the set of teachers at the school. We may consider relations on P of the type a “is a teacher of” d . For example, if a is a teacher of d, e, f, g ; b is a teacher of h, i ; and c is a teacher of k, l , we use R to mean “is a teacher of” and write the relations

$$a R d, a R e, a R f, a R g$$

$$b R h, b R i, c R k, c R l$$

$\square$

Example 1.6.4

Let $X = \{2, 3, 4, 5\}$ and R mean “is divisible by in $\mathbb{N}$.” Then

$$\begin{aligned} X \times X = \{ & (2, 2), (2, 3), (2, 4), (2, 5), (3, 2), (3, 3), (3, 4), (3, 5), \\ & (4, 2), (4, 3), (4, 4), (4, 5), (5, 2), (5, 3), (5, 4), (5, 5) \} \end{aligned}$$

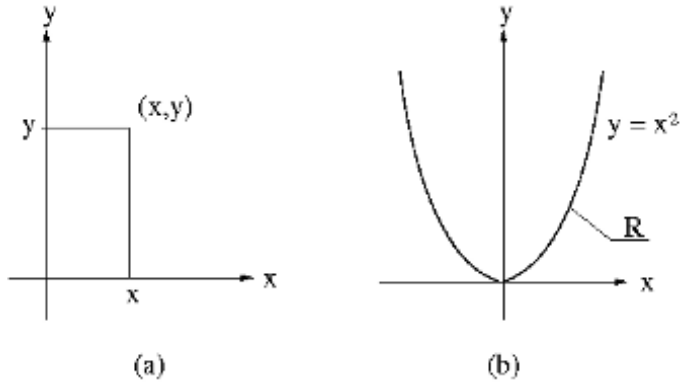
Then $2 R 2, 3 R 3, 4 R 2, 4 R 4$, and $5 R 5$; i.e.,

$$R = \{(2, 2), (3, 3), (4, 2), (4, 4), (5, 5)\}$$

$\square$

Example 1.6.5

Let $\mathbb{R}$ denote the set of real numbers and $\mathbb{R} \times \mathbb{R}$ the set of ordered pairs (x, y) of real numbers. Descartes exploited the fact that $\mathbb{R} \times \mathbb{R}$ could be represented geometrically as a plane, with an origin $(0, 0)$ and each element $(x, y) \in \mathbb{R} \times \mathbb{R}$ a point with *Cartesian coordinates* (x, y) measured off in perpendicular distances x and then y according to some preselected directions (the x and y coordinate axes) and some preselected scale. This is illustrated in Fig. 1.3 (a).

**Figure 1.3**

Cartesian coordinates and the relation $y = x^2$ from $\mathbb{R}$ into $\mathbb{R}$.

Now consider the relation

$$R = \{(x, y) : x, y \in \mathbb{R}, y = x^2\}$$

that is, R is the set of ordered pairs such that the second member of the pair is the square of the first. This relation, of course, corresponds to the sets of all points on the parabola. The rule $y = x^2$ simply identifies a special subset of the Cartesian product (the plane) $\mathbb{R} \times \mathbb{R}$, see Fig. 1.3 (b). $\square$

We now list several types of relations that are of special importance. Suppose R is a relation on a set A , i.e., $R \subseteq A \times A$; then R may fall into one of the following categories:

1. *Reflexive*. A relation R is *reflexive* if and only if for every $a \in A$, $(a, a) \in R$; that is, $a R a$, for every $a \in A$.
2. *Symmetric*. A relation R is *symmetric* if and only if $(a, b) \in R \implies (b, a) \in R$; that is, if $a R b$, then also $b R a$.
3. *Transitive*. A relation R is *transitive* if and only if $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$; that is, if $a R b$ and if $b R c$, then $a R c$.
4. *Antisymmetric*. A relation R is *antisymmetric* if and only if for every $(a, b) \in R$, $(b, a) \in R \implies a = b$; that is, if $a R b$ and $b R a$, then $a = b$.

The next two sections are devoted to a discussion of two fundamental classes of relations satisfying some of these properties.

Exercises

Exercise 1.6.1 Let $A = \{\alpha, \beta\}$, $B = \{a, b\}$, and $C = \{c, d\}$. Determine

- (i) $(A \times B) \cup (A \times C)$
- (ii) $A \times (B \cup C)$
- (iii) $A \times (B \cap C)$

Exercise 1.6.2 Let R be the relation $<$ from the set $A = \{1, 2, 3, 4, 5, 6\}$ to the set $B = \{1, 4, 6\}$.

- (i) Write out R as a set of ordered pairs.
- (ii) Represent R graphically as a collection of points in the xy -plane $\mathbb{R} \times \mathbb{R}$.

Exercise 1.6.3 Let R denote the relation $R = \{(a, b), (b, c), (c, b)\}$ on the set $R = \{a, b, c\}$. Determine whether or not R is (a) reflexive, (b) symmetric, or (c) transitive.

Exercise 1.6.4 Let R_1 and R_2 denote two nonempty relations on set A . Prove or disprove the following:

- (i) If R_1 and R_2 are transitive, so is $R_1 \cup R_2$.
- (ii) If R_1 and R_2 are transitive, so is $R_1 \cap R_2$.
- (iii) If R_1 and R_2 are symmetric, so is $R_1 \cup R_2$.
- (iv) If R_1 and R_2 are symmetric, so is $R_1 \cap R_2$.

1.7 Partial Orderings

Partial Ordering. One of the most important kinds of relations is that of *partial ordering*. If $R \subset A \times A$ is a relation, then R is said to be a *partial ordering* of A iff R is:

- (i) transitive
- (ii) reflexive
- (iii) antisymmetric

We also may say that A is *partially ordered* by the relation R . If, additionally, every two elements are comparable by R , i.e., for any $a, b \in A$, $(a, b) \in R$ or $(b, a) \in R$, then the partial ordering R is called the *linear ordering* (or *total ordering*) of set A and A is said to be *linearly (totally) ordered* by R .

Example 1.7.1

The simplest possible example of partial ordering is furnished by any subset A of the real line $\mathbb{R}$ and the usual $\leq$ (less than or equal) inequality relation. In fact, since every two real numbers are comparable, A is totally ordered by $\leq$. $\square$

Example 1.7.2

A nontrivial example of partial ordering may be constructed in $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$. Let $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ be two points of $\mathbb{R}^2$. We shall say that

$$\mathbf{x} \leq \mathbf{y} \quad \text{iff} \quad x_1 \leq y_1 \quad \text{and} \quad x_2 \leq y_2$$

Note that we have used the same symbol $\leq$ to define the new relation as well as to denote the usual “greater or equal” correspondence for the numbers (coordinates of $\mathbf{x}$ and $\mathbf{y}$). The reader will easily verify that the relation is transitive, reflexive, and antisymmetric, and therefore it is a partial ordering of $\mathbb{R}^2$. It is *not*, however, a total ordering of $\mathbb{R}^2$. To visualize this let us pick a point $\mathbf{x}$ in $\mathbb{R}^2$ and try to specify points $\mathbf{y}$ which are “greater or equal” (i.e., $\mathbf{x} \leq \mathbf{y}$) than $\mathbf{x}$ and those which are “smaller or equal” (i.e., $\mathbf{y} \leq \mathbf{x}$) than $\mathbf{x}$.

Drawing horizontal and vertical lines through $\mathbf{x}$ we subdivide the whole $\mathbb{R}^2$ into four quadrants (see Fig. 1.4). It is easy to see that the upper-right quadrant (including its boundary) corresponds to the points $\mathbf{y}$ such that $\mathbf{x} \leq \mathbf{y}$, while the lower-left quadrant contains all $\mathbf{y}$ such that $\mathbf{y} \leq \mathbf{x}$. In particular, all the points which do not belong to the two quadrants are neither “greater” nor “smaller” than $\mathbf{x}$. Thus $\leq$ is not a total ordering of $\mathbb{R}^2$. $\square$

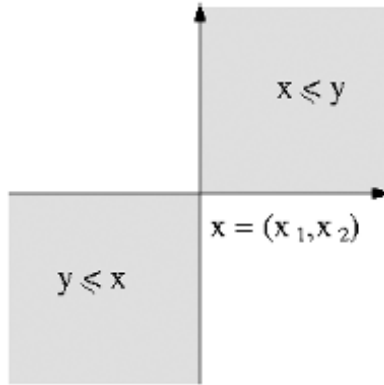
**Figure 1.4**

Illustration of a partial ordering in $\mathbb{R}$.

Example 1.7.3

Another common example of a partial ordering is furnished by the *inclusion* relation for sets. Let $\mathcal{P}(A) = 2^A$ denote the power class of the set A and define a relation R on $\mathcal{P}(A)$ such that $(C, B) \in R$ iff $C \subset B$, where $C, B \in \mathcal{P}(A)$. Then R is a partial ordering. R is transitive; if $C \subset B$ and $B \subset D$, then $C \subset D$. R is reflexive, since $C \subset C$. Finally, R is antisymmetric, for if $C \subset B$ and $B \subset C$, then $B = C$. $\square$

The notion of a partial ordering makes it possible to give a precise definition to the idea of the greatest and least elements of a set.

- (i) An element $a \in A$ is called a *least element* of A iff $a R x$ for every $x \in A$.
- (ii) An element $a \in A$ is called a *greatest element* of A iff $x R a$ for every $x \in A$.
- (iii) An element $a \in A$ is called a *minimal element* of A iff $x R a \Rightarrow x = a$ for every $x \in A$.
- (iv) An element $a \in A$ is called a *maximal element* of A iff $a R x \Rightarrow x = a$ for every $x \in A$.

While we used the term “ a is a least element,” it is easy to show that when a exists, it is unique. Indeed, if a_1 and a_2 are two least elements of A , then in particular $a_1 R a_2$ and $a_2 R a_1$ and therefore $a_1 = a_2$.

Similarly, if the greatest element exists, then it is unique. Note also, that in the case of a totally ordered set, every two elements are comparable and therefore the notions of the greatest and maximal as well as the least and minimal elements coincide with each other (a maximal element is the greatest element of all elements comparable with it). In the general case, of a set only partially ordered, if the greatest element exists, then it is also the unique maximal element of A (the same holds for the least and minimal elements). The notion of maximal (minimal) elements is more general in the sense that there may not be a greatest (least) element, but still maximal (minimal) elements, in general not unique, may exist. [Examples 1.7.4–1.7.6](#) illustrate the difference.

The notion of a partial ordering makes it also possible to define precisely the idea of *bounds* on elements of sets. Suppose R is a partial ordering on a set B and $A \subset B$. Then, continuing our list of properties, we have:

- (v) An element $b \in B$ is an *upper bound* of A iff $x R b$ for every $x \in A$.
- (vi) An element $b \in B$ is a *lower bound* of A iff $b R x$ for every $x \in A$.
- (vii) The *least upper bound* of A (i.e., the *least* element of the set of all upper bounds of A), denoted $\sup A$, is called the *supremum* of A .
- (viii) The *greatest lower bound* of A , denoted $\inf A$, is called the *infimum* of A .

Note that if A has the *greatest* element, say a , then $a = \sup A$. Similarly, the *smallest* element of a set coincides with its *infimum*.

Example 1.7.4

The set $\mathbb{R}$ of real numbers is totally ordered in the classical sense of real numbers. Let $A = [0, 1) = \{x \in \mathbb{R} : 0 \leq x < 1\} \subset \mathbb{R}$ be the interval “closed on its left end and open on the right.” Then:

- A is bounded from above by every $y \in \mathbb{R}$, $y \geq 1$.

- $\sup A = 1$.
- There is neither a greatest nor a maximal element of A .
- A is bounded from below by every $y \in \mathbb{R}$, $y \leq 0$.
- $\inf A =$ the least element of $A =$ the minimal element of $A = 0$.

□

Example 1.7.5

Let $\mathcal{Q}$ denote the set of all rational numbers and A be a subset of $\mathcal{Q}$ such that for every $a \in A$, $a^2 < 2$. Then:

- A is bounded from above by every $y \in \mathcal{Q}$, $y > 0$, $y^2 > 2$.
- There is not a least upper bound of A ($\sup A$) and therefore there is neither a greatest nor a maximal element of A .
- Similarly, no $\inf A$ exists.

□

We remark that set A is referred to as *order complete* relative to a linear ordering R if and only if every nonempty subset of A that has an upper bound also has a least upper bound. This idea makes it possible to distinguish between real numbers (which are order complete) and rational numbers (which are not order complete).

Example 1.7.6

Let $A \subset \mathbb{R}^2$ be the set represented by the shaded lower-left area in [Fig. 1.5](#), including its boundary, and consider the partial ordering of $\mathbb{R}^2$ discussed in [Example 1.7.2](#). Then:

- The first (upper-right) quadrant of $\mathbb{R}^2$, denoted B (including its boundary), consists of all *upper bounds* of A .
- The origin $(0, 0)$ is the *least* element of B and therefore is the *supremum* of A .
- All the points belonging to the “outer” corner (see [Fig. 1.5](#)) are *maximal elements* of A .
- There is not a *greatest* element of A .

□

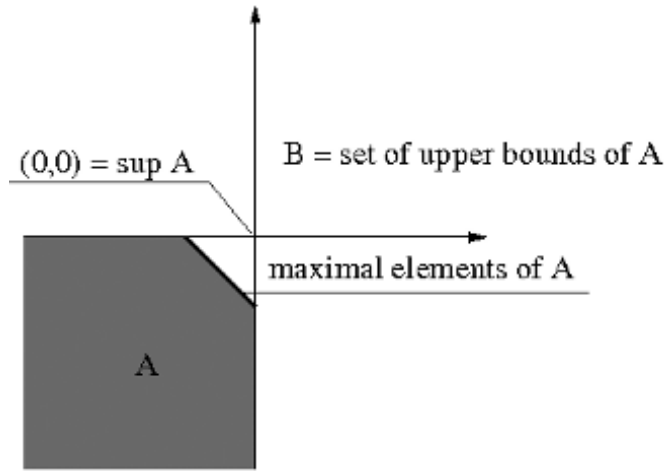


Figure 1.5

Illustration of the notion of upper bound, supremum, maximal, and greatest elements of a set.

The Kuratowski–Zorn Lemma and the Axiom of Choice. The notion of maximal elements leads to a fundamental mathematical axiom known as the Kuratowski–Zorn Lemma:

The Kuratowski–Zorn Lemma. Let A be a nonempty, partially ordered set. If every linearly ordered subset of A has an upper bound, then A contains at least one maximal element.

Kuratowski–Zorn’s Lemma asserts the existence of certain maximal elements without indicating a constructive process for finding them. It can be shown that Kuratowski–Zorn Lemma is equivalent to the axiom of choice:

Axiom of Choice. Let $\mathcal{A}$ be a collection of disjoint sets. Then there exists a set B such that $B \subset \cup A$, $A \in \mathcal{A}$ and, for every $A \in \mathcal{A}$, $B \cap A$ has exactly one element.

The Kuratowski–Zorn Lemma is an essential tool in many existence theorems covering *infinite-dimensional vector spaces* (e.g., the existence of Hamel basis, proof of the Hahn–Banach theorem, etc.).

Exercises

Exercise 1.7.1 Consider the partial ordering of $\mathbb{R}^2$ from Examples 1.7.2 and 1.7.6. Construct an example of a set A that has many minimal elements. Can such a set have the least element?

Exercise 1.7.2 Consider the following relation in $\mathbb{R}^2$:

$$x R y \quad \text{iff} \quad (x_1 < y_1 \text{ or } (x_1 = y_1 \text{ and } x_2 \leq y_2))$$

- (i) Show that R is a *linear (total) ordering* of $\mathbb{R}^2$.
- (ii) For a given point $x \in \mathbb{R}^2$ construct the set of all points y “greater than or equal to” x , i.e., $x R y$.
- (iii) Does the set A from Example 1.7.6 have the greatest element with respect to this partial ordering?

Exercise 1.7.3 Consider a contact problem for the simply supported beam shown in Fig. 1.6. The set K of

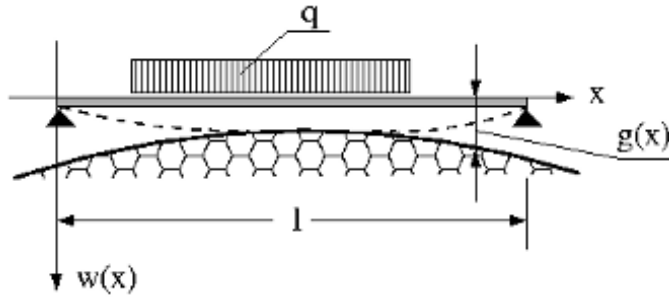


Figure 1.6

A contact problem for a beam.

all kinematically admissible deflections $w(x)$ is defined as follows:

$$K = \{w(x) : w(0) = w(l) = 0 \text{ and } w(x) \leq g(x), x \in (0, l)\}$$

where $g(x)$ is an initial gap function specifying the distance between the beam and the obstacle. Let V be a class of functions defined on $(0, l)$ including the gap function $g(x)$. For elements $w \in V$ define the relation

$$w R v (w \leq v) \text{ iff } w(x) \leq v(x) \text{ for every } x \in (0, l)$$

- (i) Show that R is a partial ordering of V .
- (ii) Show that R is not a linear ordering of V .
- (iii) Show that the set K can be rewritten in the form:

$$K = \{w(x) : w(0) = w(l) = 0 \text{ and } w \leq g\}$$

Exercise 1.7.4 Let $\mathcal{P}(A)$ denote the power class of a set A ; then $\mathcal{P}(A)$ is partially ordered by the inclusion relation (see Example 1.7.3). Does $\mathcal{P}(A)$ have the smallest and greatest elements?

1.8 Equivalence Relations, Equivalence Classes, Partitions

Equivalence Relations. A relation $R \subset A \times A$ is called an *equivalence relation* on A iff it is:

- (i) reflexive
- (ii) transitive

(iii) symmetric

Such relations may serve to generalize the familiar notion of equality ($=$) in the set of real numbers. Numerous other examples can be cited.

Example 1.8.1

Let C denote the set of male children living in a certain residential area. We may introduce a relation “is a brother of” on $C \times C$. If we accept that a given male child can be the brother of himself, then this relation on C is reflexive. Also, if a is the brother of b , then, of course, b is the brother of a . Moreover, if b is also the brother of c , then so is a . It follows that “is the brother of” is an equivalence relation on C . $\square$

Example 1.8.2

Let L denote the set of all straight lines in the Euclidean plane. The rule “is parallel to” defines a relation R on L ; indeed, R is an equivalence relation. R is transitive, since if line a is parallel to line b and b is parallel to c , then a is parallel to c ($a R b$ and $b R c \implies a R c$). R is symmetric, since $a R b$ and $b R a$, and R is reflexive if we admit that every line is parallel to itself. $\square$

Example 1.8.3

Let $\mathbb{Z}^+ = \{0, 1, 2, \dots\}$ denote the nonnegative integers and R be a relation on $\mathbb{Z}^+$ such that $(a, b) \in R$ if $a - b$ is divisible by 3 on $\mathbb{Z}$. Then R is an equivalence relation. It is reflexive, since $a - a = 0$ is divisible by 3; it is transitive, for if $a - b = 3r$ and $b - c = 3s$, where $r, s \in \mathbb{Z}$, then $a - c = 3(r + s)$. Finally, R is symmetric because $b - a$ is divisible by 3 if $a - b$ is divisible by 3. $\square$

Equivalence Classes. Let A be a set and R be an equivalence relation defined on A . If $a \in A$, the elements $x \in A$ satisfying $x R a$ constitute a subset of A , denoted $R[a]$, called an *equivalence class* of a . That is,

$$R[a] = \{x : x \in A, x R a\}$$

Example 1.8.4

Let $\mathbb{Z}^+$ be the set of nonnegative integers and define a relation R on $\mathbb{Z}^+$ such that $(a, b) \in R$ iff $a - b$ is divisible by 3 on $\mathbb{Z}$. Consider the equivalence class $R[1]$. By definition, $R[1]$ is the set of elements x in $\mathbb{Z}^+$ such that $x R 1$; i.e., $x - 1$ is divisible by 3. Hence,

$$R[1] = \{1, 4, 7, 10, \dots\}$$

Similarly,

$$R[2] = \{2, 5, 8, 11, \dots\}$$

$$R[3] = \{0, 3, 6, 9, 12, \dots\}$$

$$R[4] = \{1, 4, 7, 10, \dots\}$$

and so forth. Notice that $4 \in R[1]$ and, as a consequence, $R[4] = R[1]$. $\square$

Example 1.8.5

The rational numbers $\mathcal{Q}$ are equivalence classes on pairs of integers. Consider the relation R on $\mathcal{Z} \times (\mathcal{Z} - \{0\})$ defined by $\{(p, q), (r, s)\} \in R$ iff $ps = qr$. Then $R[(p, q)]$ is a set of pairs $(r, s) \in \mathcal{Z} \times \mathcal{Z}$ such that $(r, s)R(p, q)$; i.e., $r/s = p/q$. Of course, instead of always writing out this elaborate equivalence class notation, we prefer to simply denote

$$R[(p, q)] = \frac{p}{q}$$

$\square$

Partition of a Set. A class $\mathcal{B} \subset \mathcal{C}(A)$ of nonempty subsets of a set A is called a *partition* of A iff:

- (i) $\cup\{B : B \in \mathcal{B}\} = A$
- (ii) every pair of distinct subsets of $\mathcal{B}$ is disjoint; i.e., if $B, C \in \mathcal{B}$, $B \neq C$ then $B \cap C = \emptyset$

We are approaching the important idea of an equivalence relation R partitioning a set into equivalence classes. For example, if A is the set of all triangles, the relations “is congruent to” or “has the same area as” are equivalence relations that segregate triangles into certain equivalence classes in A . To make this assertion precise, we first need:

LEMMA 1.8.1

Let R denote an equivalence relation on a set A and $R[a]$ an equivalence class for $a \in A$. If $b \in R[a]$, then $R[b] = R[a]$.

PROOF By definition, $b \in R[a] = \{x : x \in A, xRa\} \Rightarrow bRa$; similarly, $x \in R[b] \Rightarrow xRb$. Since R is transitive, xRa , and therefore, $R[b] \subseteq R[a]$. A repetition of the argument assuming $x \in R[a]$ yields $R[a] \subseteq R[b]$, which means that $R[a] = R[b]$ and completes the proof. $\blacksquare$

Example 1.8.6

Recall [Example 1.8.4](#) in which R was a relation on $\mathcal{Z}^+$ such that $(a, b) \in R \Rightarrow a - b$ was divisible by 3. Observe that $R[1] = R[4]$. $\square$

LEMMA 1.8.2

If $R[a] \cap R[b] \neq \emptyset$, then $R[a] = R[b]$.

PROOF Suppose that $R[a] \cap R[b] = \{\alpha, \beta, \dots\} \neq \emptyset$. Then $\alpha \in R[a]$ and $\alpha \in R[b]$. By Lemma 1.8.1, this means that $R[a] = R[b]$. ■

LEMMA 1.8.3

If R is an equivalence relation on A and $R[a]$ an equivalence class for $a \in A$, then

$$\bigcup \{R[x] : x \in A\} = A$$

PROOF Let $Y = \bigcup \{R[x] : x \in A\}$. Then each $y \in Y$ belongs to an $R[x]$ for some $x \in A$, which means that $Y \subseteq A$. Consequently, $\bigcup \{R[x] : x \in A\} \subset A$. Now take $z \in A$. Since R is reflexive, $z R z$ and $z \in R[z]$. Therefore $A \subseteq \bigcup \{R[x] : x \in A\}$. This completes the proof. ■

Finally, we have:

PROPOSITION 1.8.1

An equivalence relation R on a set A effects a partitioning of A into equivalence classes. Conversely, a partitioning of A defines an equivalence relation on A .

PROOF Let $R[a], R[b], R[c], \dots$ denote equivalence classes induced on A by R , with $a, b, c, \dots \in A$. By Lemma 1.8.3, $R[a] \cup R[b] \cup R[c] \cup \dots = A$, so that property (i) of a partitioning of A is satisfied. Also (ii) is satisfied, because if $R[a]$ and $R[b]$ are distinct equivalence classes, they are disjoint by Lemma 1.8.2.

To prove the converse, let $\mathcal{B}$ be any partition of A and define a relation R on A such that $a R b$ iff there exists a set B in the partition such that $a, b \in B$. Clearly, such a relation is both reflexive and symmetric; for every $a \in A$ there is a set B in the partition such that $a \in B$. Formally, $a, a \in B$, also $a, b \in B$ implies that $b, a \in B$.

Now suppose $a R b$ and $b R c$. This implies that $a, b \in B$ for some B and $b, c \in C$ for some C . Consequently, $b \in B \cap C$, which, according to the definition of partition, implies that $B = C$. Thus both a and c belong to the same set $B = C$ and therefore $a R c$, which means that R is transitive. This completes the proof that R is an equivalence relation. ■

Quotient Sets. The collection of equivalence classes of A is a class, denoted by A/R , called the *quotient* of A by R : $A/R = \{R[a] : a \in A\}$. According to Proposition 1.8.1, the quotient set A/R is, in fact, a

partition of A .

Example 1.8.7

Let $\mathcal{S}$ be a collection of circles in $\mathbb{R}^2$, centered at the origin:

$$S(\mathbf{0}, r) = \{\mathbf{x} \in \mathbb{R}^2 : \text{distance from } \mathbf{0} \text{ to } \mathbf{x} = r\}$$

Let us identify the origin $\mathbf{0}$ with the circle of zero radius. Obviously,

$$\mathcal{S} = \{S(\mathbf{0}, r), r \geq 0\}$$

is a partition of $\mathbb{R}^2$. Defining an equivalence relation R by $\mathbf{x} R \mathbf{y}$ iff the distance from $\mathbf{x}$ to $\mathbf{0}$ = distance from $\mathbf{y}$ to $\mathbf{0}$, we may identify the circles with equivalence classes of points in $\mathbb{R}^2$ and the partition with the quotient set $\mathbb{R}^2/R$. $\square$

We shall return to the important issue of equivalence classes and quotient sets in the context of vector spaces in [Chapter 2](#).

Exercises

Exercise 1.8.1 (i) Let T be the set of all triangles in the plane $\mathbb{R}^2$. Show that “is similar to” is an equivalence relation on T .

(ii) Let P be the set of all polygons in the plane $\mathbb{R}^2$. Show that “has the same number of vertices” is an equivalence relation on P .

(iii) For part (i) describe the equivalence class $[T_0]$, where T_0 is a (unit) right, isosceles triangle with unit sides parallel to the x - and y -axes.

(iv) For part (ii) describe the equivalence class $[P_0]$, where P_0 is the unit square

$$\{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$$

(v) Specify quotient sets corresponding to the relations in (i) and (ii).

Exercise 1.8.2 Let $A = \mathbb{N} \times \mathbb{N}$, where $\mathbb{N}$ is the set of natural numbers. The relation

$$(x, y) R (u, v) \stackrel{\text{def}}{\iff} x + v = u + y$$

is an equivalence relation on A . Determine the equivalence classes $[(1, 1)]$, $[(2, 4)]$, $[(3, 6)]$.

Exercise 1.8.3 Let $X_\iota, \iota \in I$, be a partition of a set X . Show that the relation

$$x \sim y \stackrel{\text{def}}{\iff} \exists \iota \in I : x \in X_\iota \text{ and } y \in X_\iota$$

is an equivalence relation on X .

Exercise 1.8.4 Let $\sim$ be an equivalence relation on a set X . Consider the corresponding quotient set $X|_{\sim}$, i.e., the partition of X into equivalence classes $[x]_{\sim}$ corresponding to relation $\sim$. Let $\approx$ be a (potentially new, different) relation corresponding to the partition discussed in [Exercise 1.8.3](#). Demonstrate that the two equivalence relations are identical, i.e.,

$$x \sim y \iff x \approx y$$

Exercise 1.8.5 Let $X_{\iota}, \iota \in I$ be a partition of a set X . Let $\sim$ be the corresponding induced equivalence relation defined in [Exercise 1.8.3](#). Consider the corresponding (potentially different) partition of X into equivalence classes with respect to the relation $\sim$. Prove that the two partitions are identical.

Functions

1.9 Fundamental Definitions

Functions. A *function* from a set A into a set B , denoted $f : A \rightarrow B$, is a relation $f \subset A \times B$ such that:

- (i) for every $x \in A$ there exists a $y \in B$ such that $x f y$
- (ii) for every $x \in A$ and $y_1, y_2 \in B$, if $x f y_1$ and $x f y_2$, then $y_1 = y_2$

In other words, for every $x \in A$ there exists a *unique* $y \in B$ such that $x f y$. We write:

$$y = f(x)$$

If $f : A \rightarrow B$, A is called the *domain* of f , denoted $\text{dom } f$, and B is called the *codomain* of f . Clearly, a function f from A into B is a relation $R \subset A \times B$ such that for every $(a, b) \in R$, each first component a appears only once. Thus a function may be thought of as a “single-valued relation” since each element in $\text{dom } f$ occurs only once in f . We also note that $\text{dom } f \neq \emptyset$. The element $y \in B$ in $f(x) = y$ is called the *image* of $x \in A$, or the *value of the function at x* .

The *range of a function* $f : A \rightarrow B$, denoted $\mathcal{R}(f)$, is the set of elements in B that are images of elements in A ; i.e., $\mathcal{R}(f)$ is the set of all images of f :

$$\mathcal{R}(f) = \{f(a) : a \in A\}$$

$\mathcal{R}(f)$ is also sometimes called the *image set*.

Once the relations are identified with subsets of the appropriate Cartesian products, functions are identified with their *graphs*. The *graph* of function $f : A \rightarrow B$ is the set

$$\text{graph } f = \{(x, f(x)) : x \in A\}$$

It is customary to use the terms *function*, *mapping*, *transformation*, and *operator* synonymously. Thus, if $f : A \rightarrow B$, we say that “ f maps A into B ” or “ f is a transformation from A into B ” or “ f is an operator from A into B ” (in some works the term “operator” is reserved for functions whose domains are subsets of *spaces*; we consider these in [Chapter 5](#)).

Clearly, this generalization of the elementary idea of a function is in complete accord with our first notions of functions; each pair $(a, b) \in f$ associates an $a \in A$ with an element $b \in B$. The function thereby establishes a correspondence between elements of A and those of B that appear in f .

For example, we have often encountered expressions of the form

$$y = f(x) = x^2$$

which we read as “ y is a function of x .” Technically, we consider a set of (say) real numbers $\mathbb{R}$ to which there belong the elements x , and another set $\mathbb{R}^+$ of nonnegative numbers $y \in \mathbb{R}^+$. The particular subset of $\mathbb{R} \times \mathbb{R}^+$, which is the function f under consideration, is identified by the *rule* $y = x^2$. We may define the *function* f by:

$$f = \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}^+, y = x^2\}$$

Example 1.9.1

Let $\mathbb{R}$ be the real numbers and consider the relation

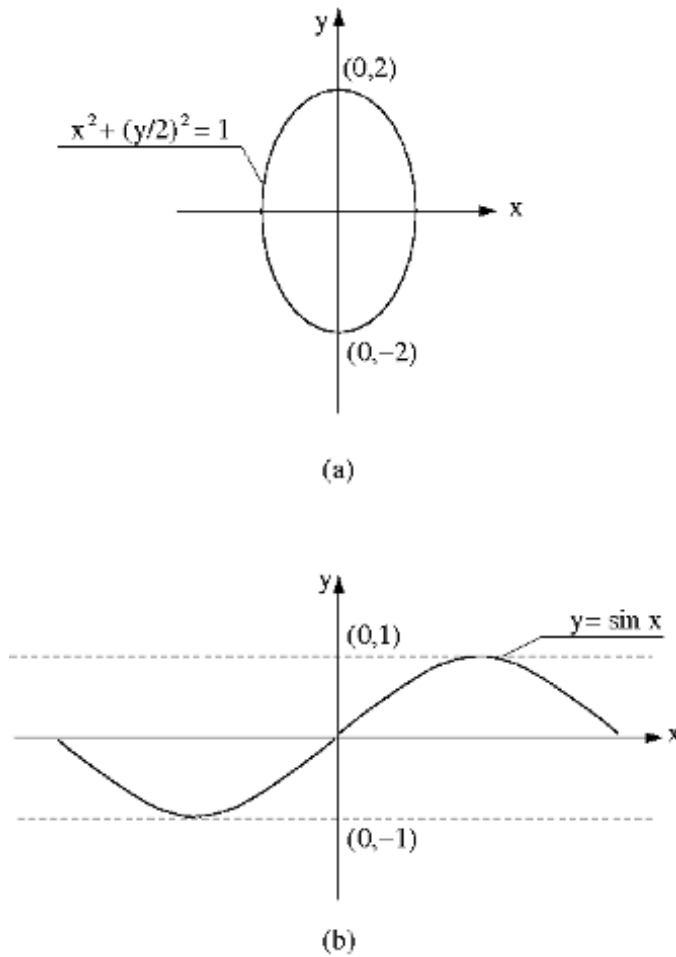
$$R = \{(x, y) : x, y \in \mathbb{R}, x^2 + \left(\frac{y}{2}\right)^2 = 1\}$$

Obviously, R defines the points on the ellipse shown in [Fig. 1.7 \(a\)](#). Notably, R is *not* a function, since elements $x \in \mathbb{R}$ are associated with *pairs* of elements in $\mathbb{R}$. For example, both $(0, +2)$ and $(0, -2) \in R$. $\square$

Example 1.9.2

The relation $R = \{(x, y) : x, y \in \mathbb{R}, y = \sin x\}$ is shown in [Fig. 1.7 \(b\)](#). This relation *is* a function. Its domain is $\mathbb{R}$, the entire x -axis, $-\infty < x < \infty$. Its codomain is also $\mathbb{R}$, i.e., the y -axis. Its range is the set $\{y : y \in \mathbb{R}, -1 \leq y \leq 1\}$. Notice that specific values of $y \in \mathcal{R}(R)$ are the images of infinitely many points in the domain of R . Indeed, $y = 1$ is the image of $\pi/2, 5\pi/2, 9\pi/2, \dots$ $\square$

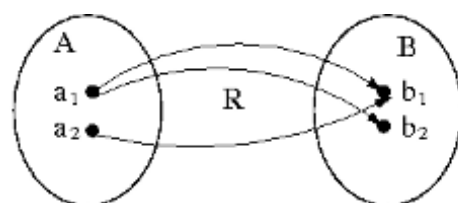
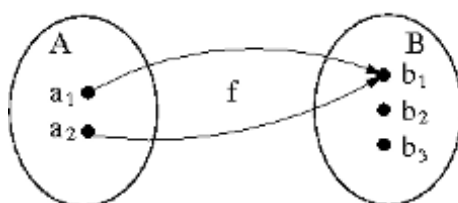
An arbitrary function $f : A \rightarrow B$ is said to map A into B and this terminology suggests nothing special about the range of f or the nature of its values in B . To identify special properties of f , we use the special nomenclature listed below:

**Figure 1.7**

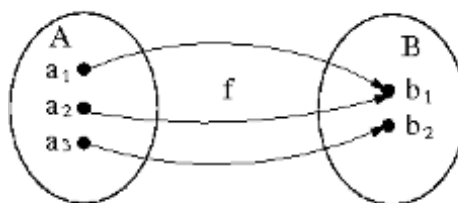
Examples of two relations on $\mathbb{R}$ of which one (a) is not a function and one (b) is a function.

1. *Surjective (Onto) Functions.* A function $f : A \rightarrow B$ is *surjective*, or from A *onto* B , if every $b \in B$ is the image of some element of A .
2. *Injective (One-to-One) Functions.* A function $f : A \rightarrow B$ is said to be *injective* or *one-to-one* (denoted 1:1) from A into B iff, for every $b \in \mathcal{R}(f)$, there is exactly one $a \in A$ such that $b = f(a)$.
3. *Bijjective (One-to-One and Onto) Functions.* A function $f : A \rightarrow B$ is *bijjective*, or *one-to-one and onto*, iff it is both injective and surjective, i.e., iff every $b \in B$ is the unique image of some $a \in A$.

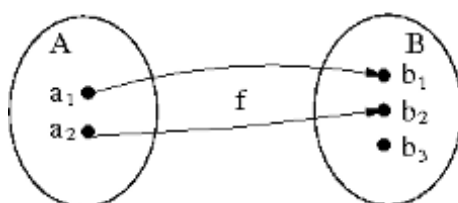
Figure 1.8 illustrates geometrically the general types of functions. The correspondence indicated in Fig. 1.8 (a) is a relation, but is not a function, because elements of A do not have distinct images in B . The one in Fig. 1.8 (d) is one-to-one, but not onto, because the element b_3 is not an image of an element of A .

(a) A relation R 

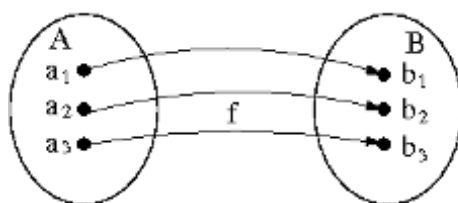
(b) Function



(c) Surjection (onto)



(d) Injection (one-to-one)



(e) Bijection (one-to-one and onto)

Figure 1.8Classification of functions $f : A \rightarrow B$.

Example 1.9.3

Let $\mathbb{R}$ denote the set of real numbers and $\mathbb{R}^+$ the set of nonnegative real numbers. Let f denote the rule $f(x) = x^2$. Then consider the following functions:

1. $f_1 : \mathbb{R} \rightarrow \mathbb{R}$. This function is not one-to-one, since both $-x$ and $+x$ are mapped into x^2 . It is not onto, since the negative real numbers are in the codomain $\mathbb{R}$, but are not images.
2. $f_2 : \mathbb{R} \rightarrow \mathbb{R}^+$. This function is not one-to-one, but it is onto.
3. $f_3 : \mathbb{R}^+ \rightarrow \mathbb{R}$. This function is one-to-one, but it is *not* onto.
4. $f_4 : \mathbb{R}^+ \rightarrow \mathbb{R}^+$. This function is bijective; it is both one-to-one and onto.

Note that although the rule $f(x) = x^2$ defining each function f_1 , f_2 , f_3 , and f_4 is the same, the four are quite different functions. $\square$

Direct and Inverse Images. The set

$$f(C) = \{f(a) : a \in C \subseteq A\}$$

is called the *direct image* of C . Obviously,

$$f(C) \subset \mathcal{R}(f)$$

Likewise, suppose $f : A \rightarrow B$ and D is a subset of B . Then the set

$$f^{-1}(D) = \{a : f(a) \in D\}$$

is called the *inverse image* of D under f . Clearly,

$$f^{-1}(D) \subset A$$

i.e., $f^{-1}(D)$ is a subset of the domain of f .

These ideas are illustrated symbolically in [Fig. 1.9](#), where the set G denotes the intersection of $\mathcal{R}(f)$ and a subset $D \subset B$. It is clear that $f^{-1}(D)$ consists of those elements in A that have images in $G \subset D$; in other words, not all of D need consist of images of elements of A . We now list several properties involving functions. Let $f : X \rightarrow Y$, $A \subset X$, $B \subset X$, $D \subset Y$, and $F \subset Y$. Then the following hold:

1. $f(A \cup B) = f(A) \cup f(B)$
2. $f(A \cap B) \subset f(A) \cap f(B)$
3. $f^{-1}(D \cup F) = f^{-1}(D) \cup f^{-1}(F)$
4. $f^{-1}(D \cap F) = f^{-1}(D) \cap f^{-1}(F)$

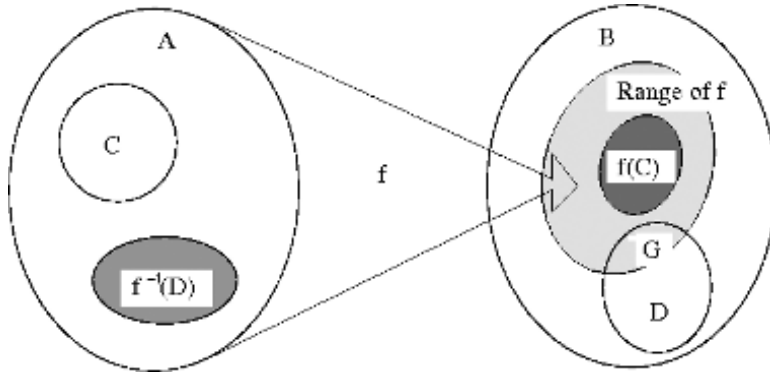
**Figure 1.9**

Illustration of $f(C)$, $\mathcal{R}(f)$, and $f^{-1}(D)$ for $f : A \rightarrow B$.

Example 1.9.4

Let $A = \{-1, -2\}$, $B = \{1, 2\}$, $f(x) = x^2$. Then:

$$A \cap B = \emptyset$$

$$f(A \cap B) = \emptyset$$

However,

$$f(A) = \{1, 4\}, \quad f(B) = \{1, 4\}$$

Consequently,

$$\begin{aligned} f(A) \cap f(B) &= \{1, 4\} \neq \emptyset \\ &\neq f(A \cap B) \end{aligned}$$

However, $f(A \cup B) = f(A) \cup f(B)$ always. $\square$

Example 1.9.5

(Proof of $f(A \cup B) = f(A) \cup f(B)$)

Let $y \in f(A \cup B)$. Then there exists an $x \in A$ or B such that $y = f(x)$. If $x \in A$, then $y = f(x) \in f(A)$; if $x \in B$, then $y = f(x) \in f(B)$. Consequently, $y \in f(A) \cup f(B)$, which proves that $f(A \cup B) \subseteq f(A) \cup f(B)$. Conversely, let $w \in f(A) \cup f(B)$. Then w is the image of an $x \in A$ or an $x \in B$; i.e., $w = f(x)$, $x \in A \cup B$. Hence $w \in f(A \cup B)$ and, therefore, $f(A) \cup f(B) \subseteq f(A \cup B)$. $\square$

Example 1.9.6

(Proof of $f^{-1}(D \cup F) = f^{-1}(D) \cup f^{-1}(F)$)

Suppose that $x \in f^{-1}(D \cup F)$. Then there exists a $y \in D$ or F such that $y = f(x)$; i.e., $f(x) \in D \cup F$. If $f(x) \in D$, $x \in f^{-1}(D)$ and if $f(x) \in F$, $x \in f^{-1}(F)$, so that $x \in f^{-1}(D) \cup f^{-1}(F)$. $\square$